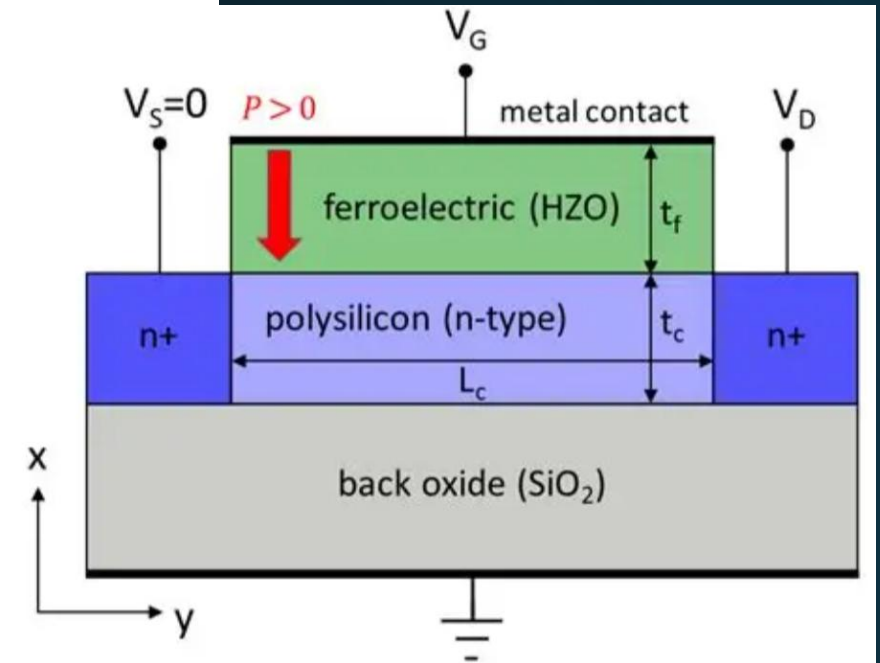


“Simulation-Based Optimization of FeFET Endurance for In-Memory Computing Applications”

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Endurance-Aware Multi-Level FeFET for In-Memory Computing

- FeFET (Ferroelectric Field Effect Transistor) is a promising non-volatile memory device combining fast operation and low power with CMOS compatibility.
- Uses a ferroelectric HfZrO_2 (HZO) layer that retains polarization even without power enabling data storage directly inside transistors.
- Enables In-Memory Computing (IMC) for AI accelerators by storing analog weights and performing Multiply–Accumulate (MAC) operations directly in memory.
- **Current Challenge:** FeFET endurance is limited ($\sim 10^5$ – 10^6 write cycles) due to charge trapping and polarization fatigue, which cause drift in multi-level states.
- **Goal:** Explore pulse-based write strategies to improve endurance while maintaining multi-bit stability.



Problem Statement & Objective

PROBLEM	• FeFET endurance degrades due to charge trapping and polarization fatigue in the ferroelectric HfZrO₂ layer. • Multi-level states (L0–L3) become unstable after repeated write cycles (level drift + overlap). • Single strong pulses (S1) cause high electric field stress, accelerating trap formation. - Blind multi-pulse methods (S2) improve accuracy but waste massive energy (~49,000 units in simulation). • Limits FeFET use in analog in-memory computing where weights require frequent updates.
OBJECTIVE	• Develop a pulse-engineering strategy that reduces device stress. • Use small incremental pulses, read-verify-stop, and rest time to allow partial trap relaxation. • Maintain 4 distinct levels (L0–L3) across ≥1000 cycles without collapse. • Reduce energy consumption compared to traditional multi-pulse methods. • Evaluate the strategy via custom endurance simulation with trap dynamics
EXPECTED OUTCOME	• Stable multi-level FeFET programming for 1000 cycles with no window collapse. • 10× lower energy vs blind multi-pulse method (S3 ≈ 5,000 vs S2 ≈ 49,000). • Moderate verify count (≈16 instead of 50), proving efficient programming. – • Validation that pulse engineering + verify + rest is an effective endurance-aware solution. • Provides a physically-plausible method for real FeFET operation and research testing. Github: https://github.com/preethi-github-29/fefet-results

Approach: Modeling, Pulse Strategies, and Simulation Workflow

Section	Details
1. Device Modeling	• Represent FeFET behavior using a compact behavioral model. • Polarization change modeled via a smooth activation function (f_{program}). • Trap accumulation modeled through stress-dependent growth ($k_{\text{stress}} \times V_{\text{pulse}}^{\alpha} \times \text{width}^{\beta}$). • Trap relaxation modeled via exponential decay ($\exp(-t_{\text{rest}} / \tau_{\text{relax}})$). Corresponds to: $f_{\text{program}}()$, $\text{update_trap}()$ in all scripts.
2. Programming Strategies Implemented	• S1 – Single Strong Pulse: One-shot write, no verify. • S2 – Multi-Pulse (No Verify): Many small pulses blindly applied. • S3 – Pulse + Verify + Rest (Proposed): Gentle pulses, read-verify-stop logic, and rest time for trap relaxation. Corresponds to: logic inside $\text{write_with_verify}()$ in all scripts.
3. Endurance Simulation	• Simulate 1000 programming cycles, rotating targets through $L0 \rightarrow L1 \rightarrow L2 \rightarrow L3$. • Apply programming strategy each cycle and update device level G, trap state, verify count, energy. • Track the last successful level for each state. Corresponds to: $\text{simulate_cycles}()$ in $v1/v2/v3$.

Approach: Modeling, Pulse Strategies, and Simulation Workflow

Section	Details
4. Failure Detection Method	• Every 50 cycles, check whether the 4 levels (L0–L3) remain separated by $\geq 20\%$ of nominal spacing. • In v3, use non-destructive checking to avoid artificially wearing the device during testing. Corresponds to: <code>level_spacing_ok()</code> + non-destructive check in <code>simulate_cycles()</code> (v3).
5. Metrics Collected	• Cycles survived (endurance). • Verify count per cycle (efficiency). • Total cumulative energy (write cost). • Stored level vs. cycle (stability). Corresponds to: printed outputs + plotting in v3 (<code>plot_run</code>).
6. Comparative Evaluation	• Run S1, S2, S3 for 1000 cycles each. • Compare pass/fail, energy use, verify count. • Determine which strategy best balances endurance and energy. Corresponds to: <code>run_and_report()</code> in v3.

Results: Endurance, Energy, and Verify Count Across Programming Strategies

Metric	S1	S2	S3 (Implemented this method)
Cycles survived	1000	1000	1000
Mean verify count	1	50	16.39
Total energy	4000	49000	5057.24
Stability of levels	Stable	Stable	Stable (despite noise)
Energy efficiency	Good	Very poor	Good
Endurance	Good	Good	Best trade-off

Discussion: Why S3 Works and How It Improves Endurance

Aspect	Before (v1, v2)	After (v3 – Final)	Aspect
Endurance	~100 to 400 cycles	1000 cycles stable	Endurance
Energy usage	High or unstable	Optimized and low	Energy usage
Level check method	Destructive (damaged device)	Non-destructive measurement	Level check method
Pulse style	Strong or blind	Gentle, precise, adaptive	Pulse style
Algorithm intelligence	None	Verify-and-stop + rest	Algorithm intelligence
FeFET realism	Missing	Improved trap & relaxation behavior	FeFET realism

Real-World Implementation & Physical Meaning

What Simulation Parameters Represent	k_{stress} → controls how quickly defects or traps accumulate inside the ferroelectric HfZrO_2 layer.	τ_{relax} → how fast the device naturally recovers	$v_{\text{eff}}, w_{\text{eff}}$ → actual voltage pulse amplitude & width	verify_tol → real circuit read-accuracy	t_{rest} → real pause between pulses
How This Is Done in Real Labs	Pulse shaping via Arbitrary Waveform Generators	Read-verify using Parameter Analyzers/Sense Amplifiers	Controlled pulse width + amplitude + delay	Standard technique: Incremental Step Pulse Programming (ISPP)	
Why S3 Is Physically Feasible	Only modifies how the FeFET is driven	Fully reproducible using common lab equipment	Matches techniques used in Flash/ReRAM/PCM	No fabrication changes required	
Experimental Validation Path	Measure threshold voltage after each pulse	Repeat for 1000 cycles	Compare level stability, energy, endurance	Apply S3 pulses to physical FeFET	

Conclusion & Q&A

Final Results

- S3 achieved **1000 stable cycles**
- **10× lower energy** vs blind multi-pulse (S2)
- Moderate verify count (**~16**), balanced efficiency
- Corrected simulation (v3) revealed true endurance capability

Future Work

- Validate S3 on real FeFET hardware
- Extend to higher-precision multi-level states